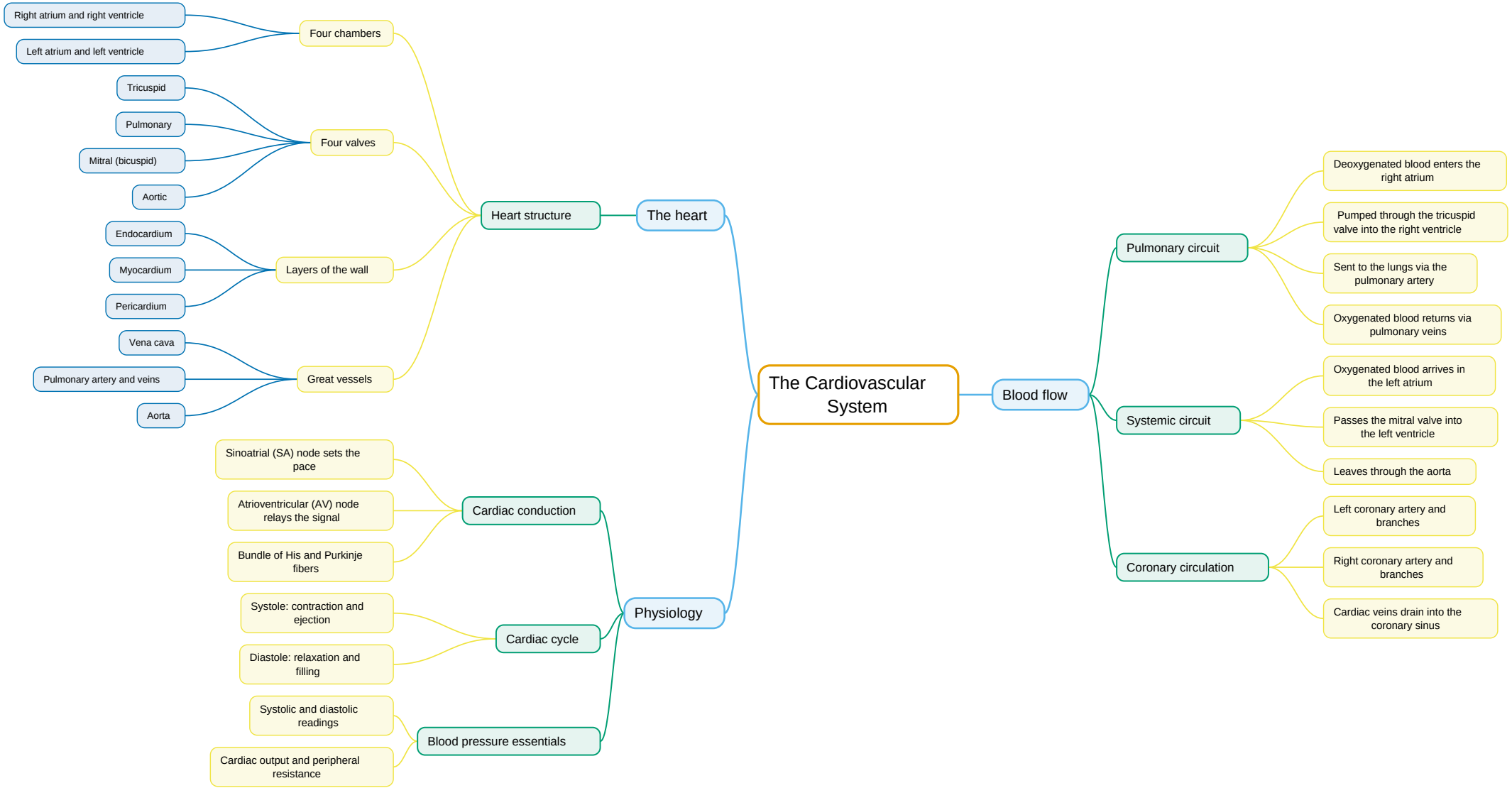
The Cardiovascular System
The heart
Heart structure
Four chambers
Right atrium and right ventricle
Left atrium and left ventricle
Four valves
Tricuspid
Pulmonary
Mitral (bicuspid)
Aortic
Layers of the wall
Endocardium
Myocardium
Pericardium
Great vessels
Vena cava
Pulmonary artery and veins
Aorta
Physiology
Cardiac conduction
Sinoatrial (SA) node sets the pace
Atrioventricular (AV) node relays the signal
Bundle of His and Purkinje fibers
Cardiac cycle
Systole: contraction and ejection
Diastole: relaxation and filling
Blood pressure essentials
Systolic and diastolic readings
Cardiac output and peripheral resistance
Blood flow
Pulmonary circuit
Deoxygenated blood enters the right atrium
Pumped through the tricuspid valve into the right ventricle
Sent to the lungs via the pulmonary artery
Oxygenated blood returns via pulmonary veins
Systemic circuit
Oxygenated blood arrives in the left atrium
Passes the mitral valve into the left ventricle
Leaves through the aorta
Coronary circulation
Left coronary artery and branches
Right coronary artery and branches
Cardiac veins drain into the coronary sinus